Web Development Track • Phase 2 • Lecture 09

CSS Selectors & Coordinate Positioning

Targeting DOM elements with precision and mastering relative, absolute, fixed, and sticky coordinate flows.

Combinator Precision Pseudo-Class States 5 Position Schemes Live Lab Build

TARGETING FUNDAMENTALS

Class (.class) vs. ID (#id) Selectors

Class Selector (.card)

Modular design building block.

Reusability: Applied across unlimited elements

Composition: Multiple classes on one tag

Specificity: Manageable weight (0, 1, 0)

```
.btn-primary { background: #2563eb; }
```

ID Selector (#navbar)

Strictly unique element identifier.

Singleton: Exactly one per document

High Specificity: Heavy weight (1, 0, 0)

Best Use: Form links and anchor jumps

```
#main-nav { border-bottom: 2px solid; }
```

RELATIONSHIP QUERIES

Descendant vs. Direct Child Combinators

SPACE SEPARATOR

Descendant Selector

Matches any nested target at any depth.

```
/* Targets anywhere inside */ nav a { color: #2563eb; }
```

Deep inheritance coverage.

GREATER-THAN SYMBOL

Direct Child Selector (>)

Matches only immediate first-level children.

```
/* Targets only immediate of */ ul > li { border-bottom: 1px solid #e2e8f0; }
```

Prevents accidental nested submenu leaks.

DYNAMIC USER STATES

Interactive Pseudo-Classes

:hover

```
button:hover { background: #1d4ed8; }
```

Mouse cursor hovering above.

:active

```
button:active { transform: scale(0.98); }
```

Physical click press down.

:focus

```
input:focus { border-color: #2563eb; }
```

Input cursor focus active.

:focus-visible

```
a:focus-visible { outline: 2px solid; }
```

Keyboard tabbed navigation outline.

MATHEMATICAL SELECTORS

Structural Power: :nth-child(n)

`:nth-child(even) & (odd)`

Zebra-stripe alternating data rows.

`:first-child & :last-child`

Target boundary items without classes.

`:nth-child(3n + 1)`

Target modular recurring grid patterns.

```
/* Modern zebra striped table */ tr:nth-child(even) { background-color: #f8fafc; } /* Clear border on last list item */ ul > li:last-child { border-bottom: none; }
```

LAYOUT MECHANICS

The 5 CSS Position Schemes

DEFAULT

static

Natural document flow. Coordinate offsets ignored.

NUDGE

relative

Offset from self. Creates parent anchor context.

DETACH

absolute

Removed from flow. Floats inside positioned parent.

SCREEN

fixed

Pinned to viewport window. Immune to scrolling.

CORE ENGINEERING PATTERN

The Relative Parent + Absolute Child Pair

1. Establish Anchor Context

Parent element declared with position: relative.

2. Float Child Into Corner

Child locked to top, right, bottom, or left offsets.

Without relative parent, absolute escapes to body edge!

```
/* 1. Parent card establishes anchor */ .card { position: relative; padding: 24px; } /* 2. Badge pinned to top-right corner */ .card .badge { position: absolute; top: 12px; right: 12px; }
```

SCROLL BEHAVIORS

Fixed vs. Sticky In Production

VIEWPORT PINNED

position: fixed;

Glued to the browser window screen.

Scroll Immunity: Never leaves view

Flow Removal: Content scrolls behind

Use Cases: Sticky navbars, modal overlays

HYBRID SCROLL THRESHOLD

position: sticky;

Scrolls naturally until threshold edge.

Parent Boundary: Stops when container ends

Requirement: Needs directional threshold (top: 0)

Use Cases: Table headers, sidebar indexes

WRAP-UP & PAIRING ACTIVITY

Session 09 Complete!

Core Takeaways

- Classes for reusable UI styling
- Direct child (>) stops style leaks
- :hover and :focus state affordances
- Relative parent anchors absolute child

Live Pairing Lab

Build a product card with a floating "SALE" badge.

Features: Relative card container, absolute badge pinned at top: 12px, button :hover transition, and sticky header.

Next Lecture: Modern Layouts: Flexbox Deep Dive

See you tomorrow!